



PowerChain | CRISIS

Full Engineering & Operations Documentation

Crisis Capital Network · Solana Runtime · Provider Resolution · Security · Operations

Version 1.0.0
29 August 2026

CORE OPERATING RULE

Mainnet means live. Missing authoritative data is unavailable or degraded - never zeroed, guessed, or silently replaced by demo data.

DOCUMENT GUIDE

How to use this handbook

A consolidated reference for engineering, operations, security and integration teams.

Audience	Start here
Developer	Quick Start -> API -> SDK -> Testing
SRE / Operations	Architecture -> Providers -> Runbook -> Troubleshooting
Security	Security Model -> Web Proxy -> Production Gate
Solana engineer	Runtime -> Program Verification -> Token Inspection -> Market Data
Product / Partner	Architecture -> PWRC Utility -> Fees -> Cross-chain -> Glossary

Repository documentation

The canonical Markdown index is docs/README.md. This handbook summarizes and connects those documents; the Markdown files remain the easiest source for code-adjacent updates.

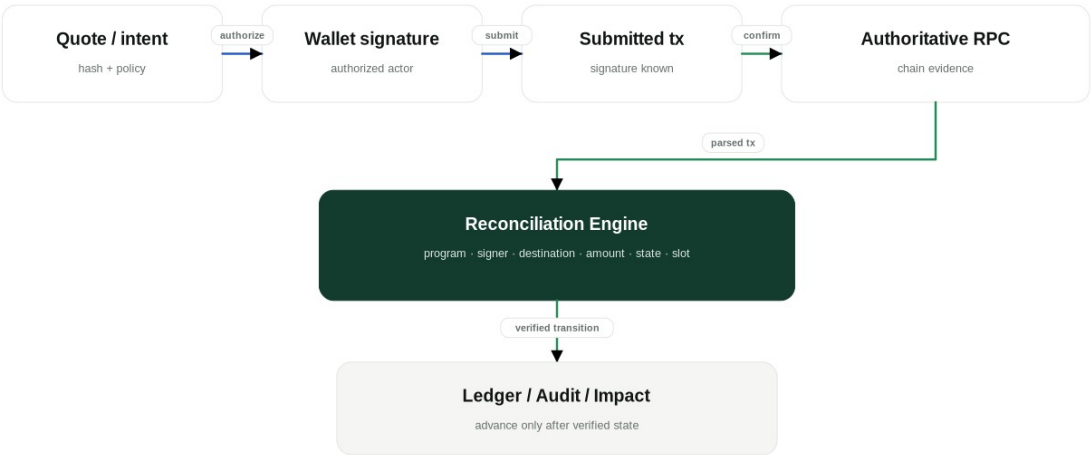
01

System architecture

One canonical service path from Solana/providers to the browser.

SOURCE OF TRUTH & RECONCILIATION

Client intent is never authoritative settlement evidence.



Source: PowerChain reconciliation architecture v1.0.0 - Unknown outcome remains EXECUTION_UNKNOWN until reconciled.

Source-of-truth architecture and authority boundaries.

```
Solana RPC / Helius + market providers
↓
@powerchain/crisis-solana-api
↓
Fastify API
↓
Next.js server-only proxy
↓
React operations UI / typed API client
```

Key boundaries

- Solana RPC is authoritative for chain state and token supply/decimals.
- Market providers contribute display/analytics observations; they cannot overwrite on-chain token truth.
- The browser talks to the website origin; private provider credentials stay in the backend.
- Compatibility aliases reuse the canonical service functions.

02

Quick start and workspace

Target toolchain, configuration and first successful requests.

Workspace	Responsibility
apps/backend	Fastify runtime and security middleware
backend	Canonical Solana service/routes/types
apps/web	Next.js proxy and operations UI
packages/api-client	Typed browser/server API client
packages/*	Policy, fees, utility, bridge, factory, rewards and chart domains
docs	Engineering/operations documentation

```
cp apps/backend/.env.example apps/backend/.env
cp apps/web/.env.example apps/web/.env.local
corepack enable
pnpm install
pnpm dev
```

TOOLCHAIN

Production releases require Node >=24.20.0, pnpm 11.24.0 and a real committed pnpm-lock.yaml. The source archive does not fabricate a lockfile.

03

HTTP API reference

System, canonical Solana and compatibility routes.

Method / Path	Purpose
GET /api/v1/health	Liveness
GET /api/v1/ready	RPC/cluster readiness
GET /api/v1/config/public	Non-secret runtime configuration
GET /api/v1/providers/status	Redacted provider telemetry
GET /api/v1/openapi.json	OpenAPI JSON
GET /api/v1/solana/overview	Cluster/RPC overview
GET /api/v1/solana/programs	Program deployment verification
GET /api/v1/solana/market?mint=	Market resolution for explicit mint
GET /api/v1/solana/assets/:mint	SPL/Token-2022 inspection
WS /api/v1/ws/solana	Bounded overview/program snapshots

Compatibility aliases

```
GET /api/solana/overview
GET /api/token/market?mint=<address>
GET /api/assets/:mint
```

Aliases emit deprecation/successor headers and call the same implementation. Only /api/token/market may use backend-configured PWRC_MINT when mint is omitted.

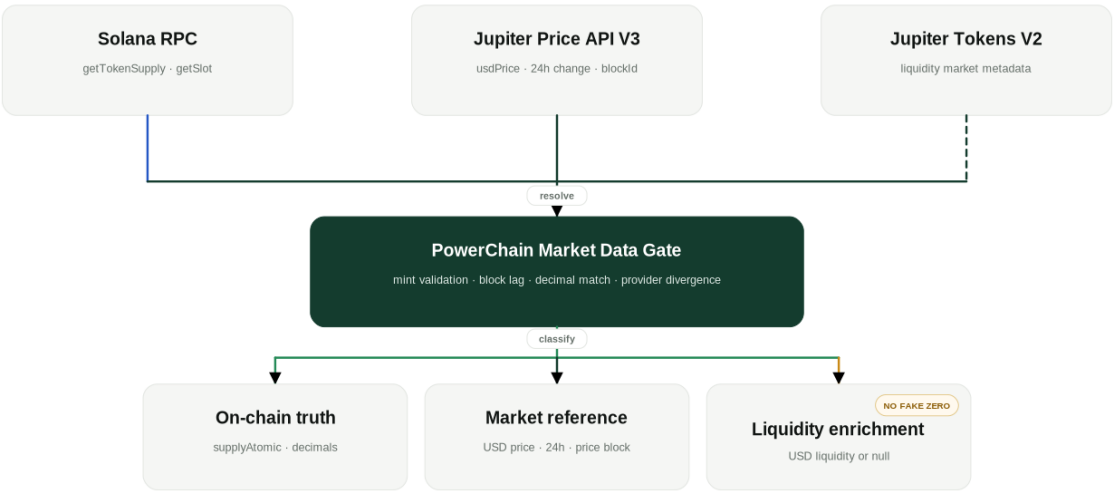
04

Solana runtime

Cluster health, genesis binding, blockhash and cache behavior.

SOLANA TOKEN MARKET DATA

On-chain supply. Jupiter market reference. Explicit provenance.



Source: PowerChain Solana data architecture v1.0.0 · Missing liquidity remains null; payment execution uses signed quotes, not display prices.

Solana market-data architecture: RPC truth plus provider observations.

RPC method	Use
getHealth	Cluster health
getSlot	Current slot
getBlockHeight	Block height
getEpochInfo	Epoch/slot index
getVersion	Solana core / feature set
getLatestBlockhash	Operational blockhash telemetry
getGenesisHash	Cluster binding
getSupply	SOL supply
getTokenSupply	Mint supply/decimals
getAccountInfo	Mint/program inspection
getMultipleAccounts	Batch program verification

CLUSTER SAFETY

SOLANA_EXPECTED_GENESIS_HASH should be configured in production. A mismatch fails readiness rather than silently accepting the RPC endpoint.

05

Launchpad and program verification

Configured IDs are inputs; executable account state is proof.

Program verification batches account inspection and distinguishes configured values from verified deployments.

State	Meaning
DEPLOYED	Exists, executable and loader accepted
NOT_FOUND	Configured account absent
UNCONFIGURED	No ID provided
INVALID_CONFIGURATION	Configured value is not a valid Solana key

Production-required Launchpad bindings

- SOLANA_LAUNCHPAD_REGISTRY_PROGRAM_ID
- SOLANA_LAUNCH_POLICY_PROGRAM_ID
- SOLANA_TOKEN_FACTORY_PROGRAM_ID

Additional vesting, escrow and Crisis program IDs can be configured without changing the verification model. Do not document or deploy guessed program IDs.

06

SPL and Token-2022 inspection

Supply, decimals, authorities and extensions are read from the mint account.

The asset endpoint calls getTokenSupply plus raw/jsonParsed account inspection. The account owner classifies classic SPL Token vs Token-2022.

Field group	Meaning
tokenProgramKind	SPL_TOKEN, TOKEN_2022 or UNKNOWN
supplyAtomic / decimals	RPC-authoritative mint supply
initialized	Mint initialization state
mintAuthority	Current mint authority or null
freezeAuthority	Current freeze authority or null
extensions[]	RPC-parsed Token-2022 extension state
metadata	Optional Helius enrichment only

AUTHORITY TRUTH

Helius metadata and market APIs never replace RPC supply, decimals, mint authority or freeze authority.

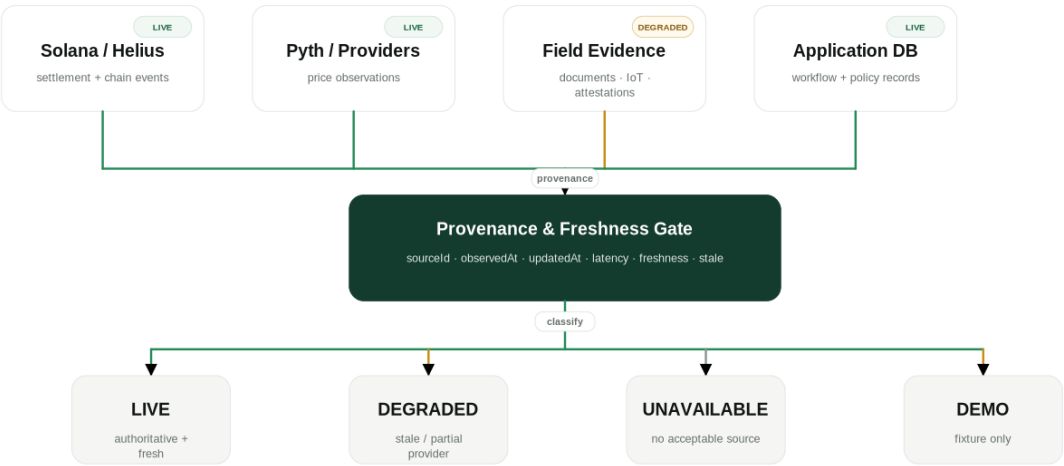
07

Market-data resolution

Source-aware price selection with divergence, freshness and confidence gates.

DATA PROVENANCE & FRESHNESS

Every operational value carries source identity and state.



Source: PowerChain data contract v1.0.0 - Mainnet renders authoritative data or explicit DEGRADED / UNAVAILABLE; demo fixtures stay labeled DEMO.

Data provenance and freshness gate.

```
Pyth
↓
Jupiter Price V3
↓
CoinGecko
↓
CoinMarketCap
↓
Birdeye
↓
Helius DAS
```

Gate	Behavior
Provider priority	First valid observation in configured order is primary
Divergence	Large cross-provider spread degrades response
Pyth confidence	Confidence/price ratio over limit degrades

Gate	Behavior
Jupiter block lag	Price block too far behind RPC slot degrades
Decimal mismatch	Jupiter vs RPC decimal mismatch degrades
No valid price	503 MARKET_DATA_UNAVAILABLE

The response explicitly marks market information as `DISPLAY_AND_ANALYTICS_ONLY`.

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Provider observability

Redacted health without leaking credentials.

`/api/v1/providers/status` exposes capability, configuration state, last latency, failure count, last check and redacted error code for operators.

Telemetry	Why it matters
configured	Distinguishes optional/unconfigured from failing
state	LIVE/DEGRADED/UNCONFIGURED/UNKNOWN
latencyMs	Detects slow upstreams
consecutiveFailures	Detects sustained provider failure
cache.entries	Shows cache footprint
cache.inFlight	Shows upstream request pressure

SECRET BOUNDARY

Provider status never returns RPC URLs, API keys, auth headers or internal tokens.

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Next.js proxy and browser security

Same-origin browser APIs without shipping provider secrets.

```
Browser -> website /api/* -> server-only proxy -> Fastify -> providers
```

- `POWERCHAIN_API_URL` is server-only.
- `POWERCHAIN_INTERNAL_API_TOKEN` is attached by the Next.js server.
- Proxy requests use no-store and a 10-second timeout.
- Proxy loops are rejected.

- Sensitive NEXT_PUBLIC_* configuration is rejected by the production gate.

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Realtime WebSocket protocol

Bounded operational snapshots, not raw RPC streaming.

WS /api/v1/ws/solana supports overview and programs topics. The server sends a hello frame, immediate snapshot and 30-second interval snapshots.

Limit	Value
Max WebSocket payload	8 KiB
Application messages per connection	30
Incoming message size check	4096 bytes
Allowed topics	overview, programs
Policy-close code	1008

AUTHORITY

Realtime snapshots are observability data. They are not settlement confirmation or treasury evidence.

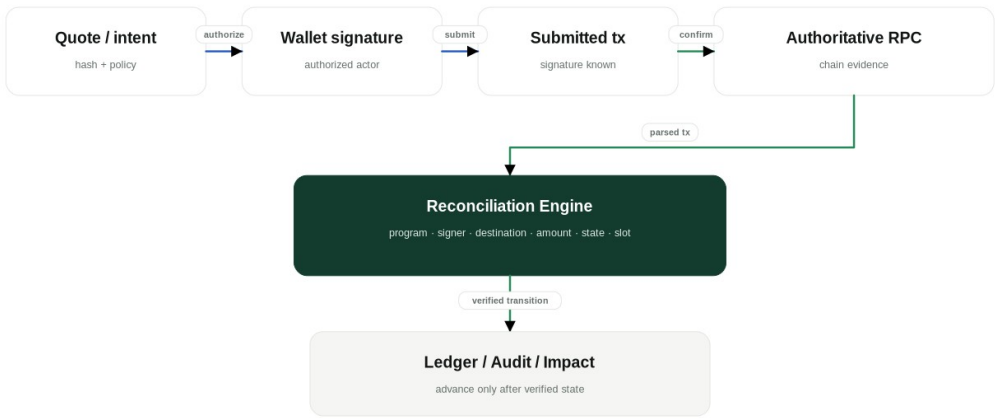
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Security model

Fail closed at every authority boundary.

SOURCE OF TRUTH & RECONCILIATION

Client intent is never authoritative settlement evidence.



Source: PowerChain reconciliation architecture v1.0.0 - Unknown outcome remains EXECUTION_UNKNOWN until reconciled.

Source-of-truth and trust-boundary view.

```
PWRC -> utility
Identity -> actor
Role -> scope
Policy -> allowed action
Approval -> authorization
Signer -> execution
Solana -> settlement evidence
Evidence -> real-world proof
```

- Constant-time internal-origin token comparison.
- Explicit CORS in production.
- Helmet + 1 MiB body limit + rate limiting.
- Request IDs and redacted logs.
- No-store API responses.
- No AI/token/provider response can grant treasury authority.

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Production deployment gate

Configuration must prove the target environment is real.

Required gate	Reason
Node >=24.20.0 / pnpm 11.24.0	Pinned toolchain
pnpm-lock.yaml	Reproducible install
mainnet-beta + dev fallback off	No public-development path
Private RPC/Helius	Production capacity

Required gate	Reason
Expected genesis hash	Cluster binding
Internal token >=32 chars	Origin authentication
Explicit CORS	Browser origin control
Required Launchpad IDs	Deployment binding
Credentialed market provider	No keyless production market dependency
No sensitive NEXT_PUBLIC_*	Browser secret prevention

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Operations runbook

What operators check every day and during incidents.

Daily: readiness -> provider status -> Solana overview -> required programs -> market degradation -> application/provider error rates.

High-priority incidents

Incident	Immediate action
RPC unavailable	Stop financial execution; validate private RPC/Helius
Genesis mismatch	Stop execution; verify cluster/config
Market divergence	Inspect observations/freshness/confidence/block lag
Required program missing	Disable affected capability; verify deployment evidence
Provider key exposure	Rotate secret, rebuild, audit browser artifacts
Unknown transaction outcome	Use EXECUTION_UNKNOWN; reconcile before retry

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Testing and release validation

Source checks, runtime smoke tests and target-environment gates.

```
pnpm check:source
pnpm typecheck
pnpm test
pnpm build
NODE_ENV=production POWERCHAIN_ENV=production pnpm check:production
```

The deterministic runtime smoke test stubs RPC/Jupiter responses and verifies overview, market, Token-2022 inspection and provider telemetry. It does not replace live devnet/mainnet integration tests.

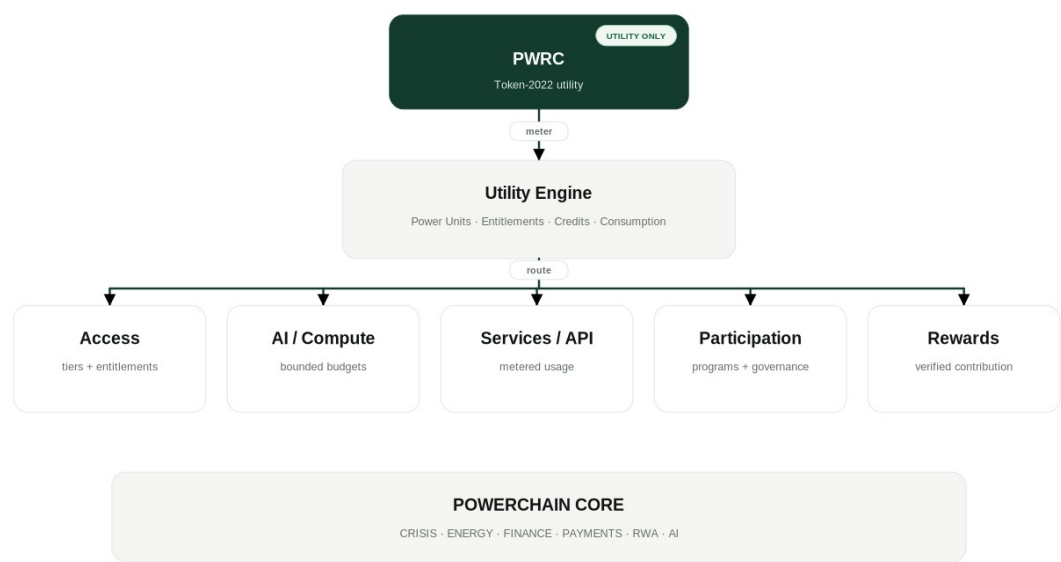
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PWRC utility and authority separation

Power Units and utility tiers are service entitlements, not RBAC.

PWRC UTILITY ARCHITECTURE · v1.0.0

Utility powers the network. Stable assets fund relief.



Source: PWRC Utility Architecture v1.0.0 · PWRC balance never grants treasury authority.

PWRC utility architecture.

Category	Examples
ACCESS	Entitlements and tiers
COMPUTE	AI workloads
SERVICES	Reports/data services
PARTICIPATION	Programs
GOVERNANCE	Economic governance
INCENTIVES	Reviewed rewards
REPUTATION	Eligibility input
NETWORK	API/service consumption

FINANCIAL BOUNDARY

PWRC balance alone never grants FUND, ALLOCATE, ESCROW, RELEASE, COMMISSION, BRIDGE, MINT or REWARD authority.

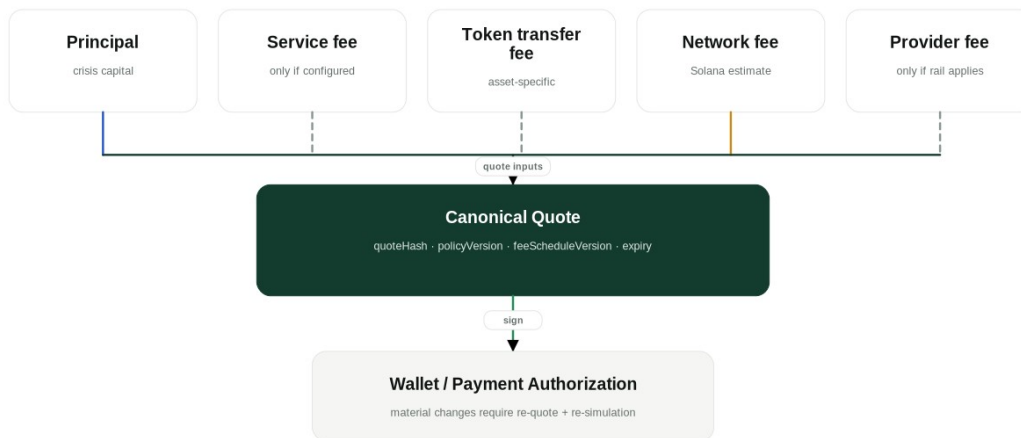
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Fees and successful-funding economics

Contributor charges and post-success commission are separate.

FEE STACK · SIGNED QUOTE

Every applicable fee is named before authorization.



Source: PowerChain fee model v1.0.0 - The 5% success commission is charged from successful pool proceeds, not contributor checkout.

Fee-stack semantics.

Contributor checkout may include principal, service fee, token transfer fee, estimated network fee and provider/onramp fee. The successful-funding commission is not charged at contributor authorization.

```

Successful pool proceeds
-> 5% total
  -> 2 percentage points Community Treasury
  -> 3 percentage points Ecosystem & Development
  
```

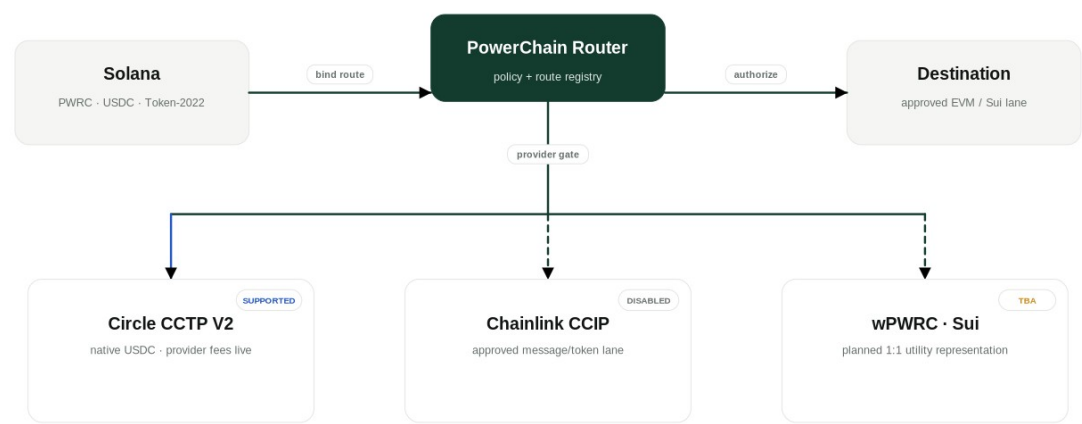
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Cross-chain security

CCTP, CCIP and wrapped-PWRC routes use different trust assumptions.

CROSS-CHAIN TRUST ROUTES

No generic bridge switch. Each route has its own controls.



Source: PowerChain cross-chain security model v1.0.0 · Route limits, replay key, finality and audit status are mandatory before execution.

Cross-chain route security model.

- Route status must be LIVE.
- Audit flag is required.
- Route ID/version and replay domain must match.
- Transfer IDs are replay-protected.
- Per-transfer and outstanding caps are enforced.
- Wrapped supply cannot exceed verified backing.

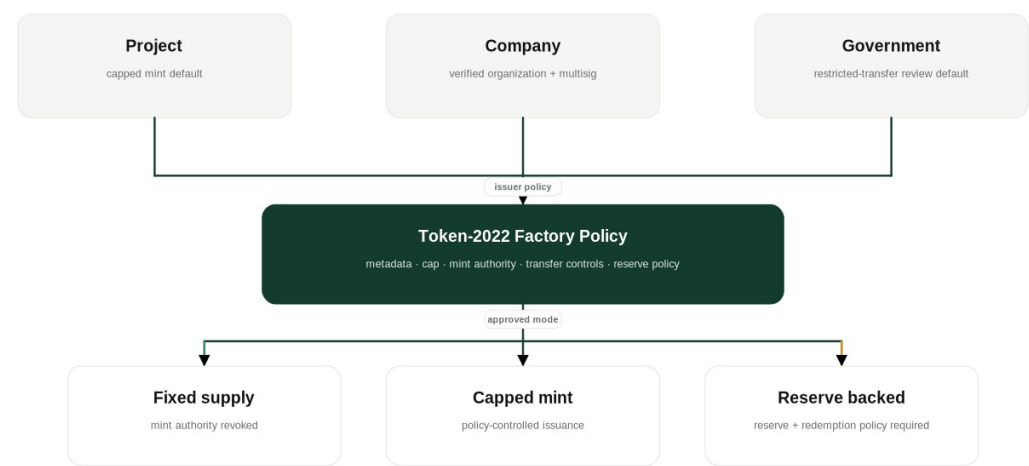
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Token factory and community rewards

Production review is not the same as automatic issuance.

TOKEN FACTORY · POLICY FIRST

One factory. Safer defaults by issuer profile.



Source: Token Factory Security v1.0.0 · Default planning cap and fee never bypass issuer, signer, reserve or compatibility review.

Token factory issuer-profile architecture.

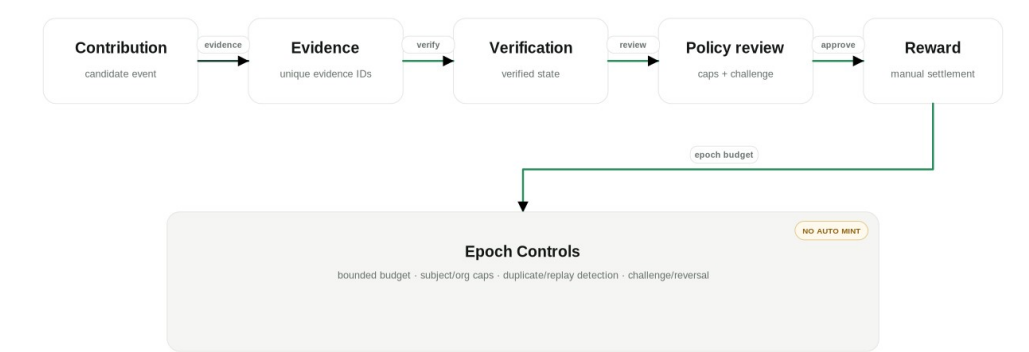
Token factory

Project, Company and Government profiles share policy validation for ticker, metadata, decimals, supply cap, reserve-backed rules, restricted-transfer review and confidential-transfer review.

Rewards

COMMUNITY REWARD SYSTEM

Reward verified contribution, not passive holding.



Source: Community Rewards roadmap v1.0.0 · Rewards never grant treasury authority or automatically change trust status.

Evidence-bound community reward flow.

Reward epochs enforce evidence, duplicate/replay prevention, subject/org caps and total epoch budget. Recommendations never auto-mint; authority remains REWARD_POLICY_REVIEW.

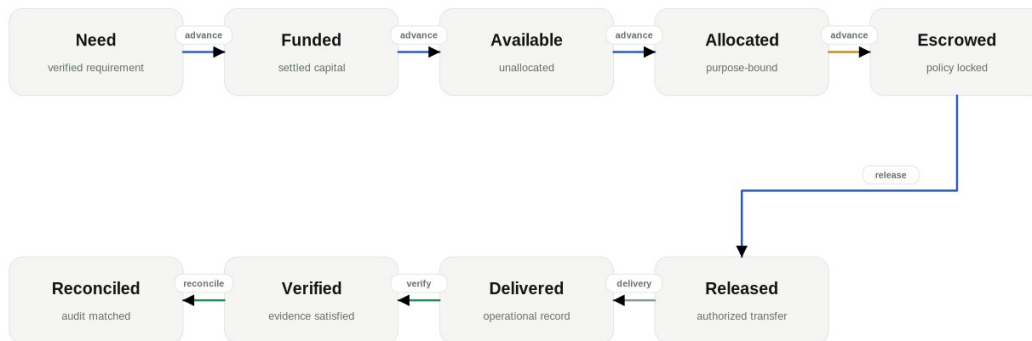
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Charts and data provenance

Every live visual should answer where, when and how trustworthy.

CAPITAL LIFECYCLE · DISTINCT STATES

Raised is not released. Released is not verified impact.



Source: CCN capital lifecycle v1.0.0 · State values must never be collapsed into one generic progress percentage.

Capital lifecycle visualization.

Charts use LIVE, DEMO, STATIC, DEGRADED, TBA and UNAVAILABLE modes. LIVE without a source becomes unavailable; stale live data becomes degraded. Missing provider data is not rendered as zero.

Architecture visual rules

- Orthogonal connectors and explicit arrowheads.
- Semantic connector labels.
- Dashed paths for optional/degraded/TBA edges.
- Status/source annotations.
- Dark forest-green cards use white text for contrast.

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SDK, Postman and integration workflow

How clients should integrate without bypassing the platform boundary.

```

const api = new PowerChainApiClient("");
await api.ready();
await api.solanaOverview();
await api.solanaPrograms();
await api.solanaMarket(mint);
await api.solanaAsset(mint);
  
```


New code should use canonical client methods. Compatibility methods exist only for migration. The Postman collection supports both backend-origin and website-origin verification.

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Troubleshooting guide

Diagnose configuration, provider, proxy and token-inspection failures.

Error	First check
POWERCHAIN_API_URL_REQUIRED	apps/web/.env.local
POWERCHAIN_API_PROXY_LOOP	Backend URL must not equal website origin
BACKEND_ORIGIN_AUTH_REQUIRED	Internal token alignment
MARKET_DATA_UNAVAILABLE	Provider credentials/priority/token support
PWRC_MINT_NOT_CONFIGURED	Legacy default only
UNSUPPORTED_TOKEN_PROGRAM	Address/cluster correctness
INVALID_CONFIGURATION	Program ID format
Lockfile missing	Generate with pinned pnpm in target environment

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Glossary and final release checklist

Common terminology and the minimum path to a production claim.

Term	Meaning
Canonical route	Supported /api/v1 contract
Compatibility alias	Deprecated route calling canonical service
Genesis binding	Expected vs active cluster genesis hash
Market observation	Provider price/liquidity/volume sample with provenance
Power Units	PWRC service-metering abstraction
EXECUTION_UNKNOWN	Transaction outcome not yet established
Deployment verification	Executable program account + accepted loader

Production claim checklist

- Frozen dependency install passes.
- Typecheck/test/build pass.
- Production gate passes.
- Readiness returns intended cluster.
- Required programs verify.
- Provider credentials and telemetry are healthy.
- Browser bundles contain no provider secrets.
- Runbooks and rollback are approved.
- Deployment evidence records commit/image/config/program IDs.

FINAL RULE

Source support is not the same as deployed capability. Production status must be backed by the target environment, not by documentation alone.